

# Basics of Cloud Computing Using Google Cloud Platform

Day 12 · Digital Science · Cloud Computing & Big Data Analytics

# Today's Agenda

- Why Should You Care About Cloud Computing?
- What Exactly IS Cloud Computing? — Service & Deployment Models
- GCP History & Philosophy
- GCP Global Infrastructure — Regions, Zones, Compliance
- GCP Compute Services — Choosing the Right Tier
- GCP Storage Services — Where Data Lives
- BigQuery — The Data Scientist's Best Friend
- GCP IAM & Security — Who Can Touch Your Data?
- GCP vs. AWS vs. Azure — Honest Comparison
- Lesson Summary & Five Things to Remember

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# Why Should You Care About Cloud Computing?

# The Cloud Revolution

- Netflix streams to 260 million subscribers simultaneously in 4K — no local servers
- A grad student can spin up 208 vCPUs, 5.7 TB RAM, 8 NVIDIA A100 GPUs in 3 minutes
- Pay only for the 4 hours your training job needs — then shut it down
- In 2005: no grant = no project
- Today: a terminal + 4 CLI commands = supercomputer access
- That is the revolution we are studying today

## **KEY TAKEAWAY — Cloud Revolution**

Cloud computing democratized access to infrastructure — you no longer need to own hardware to do world-class data science.

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# What Exactly IS Cloud Computing?

# Cloud Computing — Definition

- On-demand delivery of computing resources over the internet
- Pay-as-you-go pricing — no upfront hardware purchase
- The "cloud" = someone else's data centre
- Resources include: servers, storage, databases, networking, software, analytics, AI

# Three Service Models

| Model                 | You Manage            | Provider Manages     | Example               |
|-----------------------|-----------------------|----------------------|-----------------------|
| IaaS (Infrastructure) | OS & everything above | Hardware, networking | Compute Engine VM     |
| PaaS (Platform)       | Application code only | OS, runtime, scaling | Cloud Run, App Engine |
| SaaS (Software)       | Nothing               | Everything           | Gmail, BigQuery       |

# Four Deployment Models

- Public Cloud — Shared infrastructure (GCP, AWS, Azure). Most cost-effective for variable workloads
- Private Cloud — Dedicated infrastructure, on-premises or colo. More control, higher cost
- Hybrid Cloud — Mix of public + private. Very common with legacy on-prem systems
- Multi-Cloud — Split across providers to avoid vendor lock-in
- As data scientists: mostly Public Cloud, increasingly Multi-Cloud

## **KEY TAKEAWAY — Service & Deployment Models**

Know the difference between IaaS / PaaS / SaaS and Public / Private / Hybrid / Multi-Cloud — these define your responsibility model and cost structure.

**3.1**

# **GCP History & Philosophy**

# Google — A Data Company

- Core insight: Google is a data company that accidentally became an advertising company
- You rent the SAME infrastructure powering google.com, YouTube, Gmail, Maps
- Same data centres, Jupiter networking, Borg/Kubernetes scheduler, Colossus storage
- This is not a marketing line — it is literally true

# Why This Matters for Data Scientists

- Battle-tested at impossible scale — 8.5 billion search queries per day
- Data tools are first-class citizens — BigQuery, Dataflow, Pub/Sub have no equivalent elsewhere
  - ▶ AWS built cloud to support e-commerce
  - ▶ Azure built cloud to support Office 365
  - ▶ Google built cloud to support search-engine-scale analytics
- ML/AI deeply integrated — Google invented Transformers, TensorFlow, and TPU chips

# GCP Timeline

| Year | Event                                                    |
|------|----------------------------------------------------------|
| 2008 | App Engine launches — GCP's first public product         |
| 2011 | Cloud Storage and BigQuery (beta) released               |
| 2013 | Compute Engine exits beta — serious IaaS player          |
| 2014 | Kubernetes open-sourced — industry game-changer          |
| 2017 | Cloud Spanner — first globally-distributed relational DB |
| 2019 | Looker acquired for \$2.6B — BI ambitions                |
| 2023 | Vertex AI Gemini — GCP becomes an LLM platform           |

### KEY TAKEAWAY — GCP History

GCP's DNA is data analytics and machine learning — making it the most natural cloud platform for data scientists. 3rd largest by revenue but leads in Big Data, Kubernetes, and AI/ML.

**3.2**

# **GCP Global Infrastructure Regions, Zones & Compliance**

# Three-Level Hierarchy

- Zone — Single data centre with independent power, cooling, networking. Smallest fault-isolation unit
- Region — 3+ Zones within ~100 km. Inter-zone latency < 5 ms. Protects against single-DC failure
- Multi-Region — Data replicated across distant regions. Maximum availability + DR
- Scale (2024): 38+ Regions, 115+ Zones, 200+ countries
- Google's private backbone carries ~40% of the world's internet traffic

# Why Location Matters for Data Scientists

| Concern        | Impact                                 | Example                         |
|----------------|----------------------------------------|---------------------------------|
| Data Residency | GDPR may require EU data to stay in EU | Deploy BigQuery in europe-west1 |
| Latency        | Model-serving needs < 50 ms response   | Co-locate endpoint with users   |
| Cost           | Inter-region egress is not free        | 1 TB us→eu ≈ \$80               |

### **KEY TAKEAWAY — Infrastructure**

Region selection involves compliance (GDPR), performance (latency), and cost (egress). Always co-locate data and compute in the same region.

**3.3**

# **GCP Compute Services**

## **Picking the Right Tool**

# Compute Engine — IaaS (1/2)

## *Infrastructure as a Service — Maximum Control*

- Full VM with root access — you choose machine family, type, OS, region
- Machine Families:
  - ▶ N2 — General purpose
  - ▶ C2 — Compute-optimised
  - ▶ M2 — Memory-optimised
  - ▶ A2 — GPU / accelerator
- Per-second billing with 1-minute minimum

# Compute Engine — IaaS (2/2)

- Example type: n2-standard-16 = 16 vCPUs + 64 GB RAM
- Boot disk: Debian, Ubuntu, CentOS, Windows, Deep Learning VM images
- Use for: Custom CUDA kernels, PySpark clusters, specific OS configurations
- Cost insight: n2-standard-4 running 24/7 = ~\$136/month vs. 4 hours = \$0.76

# App Engine — PaaS

## *Platform as a Service — Managed Runtime*

- Push code → Google handles OS, scaling, load balancing, health checks
- Supports Python, Java, Go, Node.js
- No server to configure
- Use for: Lightweight ML model REST APIs, internal web tools for stakeholders

# Cloud Run — Serverless Containers

## *Containers as a Service — Most Elegant for Modern Data Scientists*

- Package code as Docker container → deploy
- Scales from ZERO to hundreds of instances in seconds
- Charges only while processing requests — zero idle cost
- Handles HTTPS, TLS certificates, domain mapping automatically
- Use for: Production ML model endpoints where traffic is bursty and unpredictable

# GKE — Google Kubernetes Engine

## *Managed K8s — For Complex Orchestration*

- Fully managed Kubernetes cluster
- Define workloads as YAML manifests; GCP handles control plane
- Integrates with Kubeflow Pipelines for ML workflow orchestration
- Use for: Complex multi-service ML platforms, MLOps pipelines
- Google invented Kubernetes — GKE is the reference implementation

## KEY TAKEAWAY — Compute Services

Don't reach for a VM for everything. Match the compute tier to your needs: IaaS (full control) → PaaS/CaaS (fast iteration) → Kubernetes (complex orchestration). Cloud's value: 4 hours of compute = \$0.76 vs. \$136/month 24/7.

**3.4**

# **GCP Storage Services Because Data Has to Live Somewhere**

# The Architectural Shift

- In cloud, storage is SEPARATE from compute — independently scaled
- Spin compute up and down without touching data
- This separation is one of the most important shifts in cloud computing

# Five Storage Primitives

| Type                 | GCP Service         | Best For                  | Example Data                   |
|----------------------|---------------------|---------------------------|--------------------------------|
| Object Store         | Cloud Storage (GCS) | Large files, any format   | CSVs, model weights, images    |
| Relational SQL       | Cloud SQL / Spanner | Structured, transactional | Customer orders, feature store |
| Wide-Column NoSQL    | Cloud Bigtable      | Time-series, low-latency  | IoT sensor readings            |
| Document NoSQL       | Firestore           | Hierarchical, flexible    | User profiles, metadata        |
| Analytical Warehouse | BigQuery            | Aggregation, ad-hoc       | Sales reports, model metrics   |

# Cloud Storage (GCS) — The Data Science Default

- Durability: 99.999999999% (eleven 9s) through redundant copies
- Scale: No upper size limit; single object up to 5 TB
- Cost: Standard tier = \$0.02/GB-month (100 GB dataset = \$2/month)
- Integration: BigQuery, Dataflow, Vertex AI, Dataproc read directly from GCS
- Think of it as a hard drive of infinite size accessible from anywhere

# Storage Classes — Trade Access for Cost

| Class    | Cost/GB-Month | Access Pattern       |
|----------|---------------|----------------------|
| Standard | \$0.020       | Daily (hot data)     |
| Nearline | \$0.010       | Monthly (backups)    |
| Coldline | \$0.004       | Quarterly (archives) |
| Archive  | \$0.0012      | Yearly (compliance)  |

## KEY TAKEAWAY — Storage Services

Cloud Storage (GCS) is your default for data science. Choose the right storage class to optimize cost — most training data is Standard, but old datasets should be lifecycled to cheaper tiers.

3.5

# BigQuery

## The Data Scientist's Best Friend

# BigQuery — Critical Distinction

- BigQuery is NOT a database. It is a query engine bolted onto a distributed file system.
- Traditional DB = OLTP — thousands of small, fast reads/writes per second
- BigQuery = OLAP — scanning billions of rows for aggregate statistics
- Example OLTP: bank balance updated when you buy coffee
- Example OLAP: total coffee purchased by every customer in Germany, by city and weekday

# Columnar Storage — Why It's Faster & Cheaper

- Row-based storage: reads ALL columns even if query needs only one → wasted I/O
- Columnar storage: each column stored separately
- SUM(amount\_usd) reads only the amount\_usd column bytes
- On a 7-column table → ~7× reduction in I/O → 7× cheaper
- Pioneered by Google's Dremel engine (research paper, 2010)

# Serverless vs. Traditional Clusters

| Traditional (Hadoop/Spark)         | BigQuery                   |
|------------------------------------|----------------------------|
| Provision cluster (10-15 min)      | Execute SQL                |
| Configure memory + parallelism     | Auto-allocates compute     |
| Monitor node failures              | Transparent infrastructure |
| Shut down after job to avoid costs | Auto-deallocates resources |
| Startup: 10-15 minutes             | Startup: ZERO              |

# BigQuery ML (BQML)

- Train ML models directly in SQL — no Python, no data export, no separate infra
- Supported models: logistic regression, k-means, random forest, XGBoost, ARIMA
- Model lives in BigQuery → call via ML.PREDICT() in SQL
- Revolutionary simplification of the ML pipeline for tabular data

# BigQuery Pricing — The Column Discipline

- Pricing: \$5 per TB scanned (on-demand)
- SELECT \* on 10 TB table = \$50 per query execution
- SELECT category, amount\_usd on same table = ~\$14 (71% cheaper!)
- Cost Trap: SELECT \* at petabyte scale becomes a genuine engineering constraint
- Rule: NEVER use SELECT \* in production — always specify columns

### KEY TAKEAWAY — BigQuery

BigQuery's columnar architecture means cost = bytes scanned. Never use `SELECT *` in production. Always specify only the columns you need — it's the difference between \$50 and \$14 per query at scale.

**3.6**

# **GCP IAM & Security**

## **Who Can Touch Your Data?**

# The Security Problem

- 10 million customer records in BigQuery. How do you ensure:
  - ▶ ETL pipeline can write but not delete?
  - ▶ Analyst can read but not drop tables?
  - ▶ Contractor loses access when contract ends?
  - ▶ Leaked credentials can be revoked in seconds?
- IAM (Identity and Access Management) solves all of this

# IAM = Three Concepts

| Concept            | Question                  | Examples                                  |
|--------------------|---------------------------|-------------------------------------------|
| Principal (Who)    | Who is requesting access? | Google Account, Service Account, Group    |
| Role (What)        | What can they do?         | bigquery.dataEditor, storage.objectViewer |
| Resource (On What) | What are they accessing?  | Project, dataset, bucket, VM              |

# Role Types & Resource Hierarchy

- Basic roles (viewer/editor/owner) — Coarse-grained. AVOID in production.
- Predefined roles — Purpose-built permission bundles. USE THESE.
- Custom roles — You define exact permissions. For strict compliance.
- Resource Hierarchy: Organisation → Folder → Project → Resources
- IAM policies INHERIT DOWNWARD — powerful but dangerous if careless

# Principle of Least Privilege

- Give every identity the MINIMUM permissions required — nothing more
- ETL Pipeline: storage.objectViewer + bigquery.dataEditor (not Owner)
- Analyst: bigquery.dataViewer — read-only (not write/delete)
- ML Trainer: bigquery.dataViewer + storage.objectCreator (not admin)
- If compromised, the blast radius is limited
- In regulated industries (finance, healthcare, govt) — this is a legal requirement

# The Service Account Key Danger

- Student downloads JSON key → commits to Git → pushes to GitHub
- Within 45 minutes: crypto-miners rack up \$2,000 in compute charges
- Google monitors GitHub for leaked keys — but damage happens fast
- Solution: Use Workload Identity Federation instead of key files
- NEVER commit credentials to version control

## KEY TAKEAWAY — IAM & Security

IAM is not optional. Apply Least Privilege to every service account. Never commit credentials to version control. Use Workload Identity Federation instead of JSON key files.

**3.7**

# **GCP vs. AWS vs. Azure**

## **The Honest Comparison**

# The Big Three

| Provider | Market Share | Strength                                  | Best For                   |
|----------|--------------|-------------------------------------------|----------------------------|
| AWS      | ~33%         | Most services (250+), largest talent pool | Highest job-market demand  |
| Azure    | ~23%         | Microsoft integration (AD, O365, .NET)    | Enterprise Microsoft shops |
| GCP      | ~11%         | BigQuery, Kubernetes, AI/ML, pricing      | Data science workflows     |

# GCP Advantages for Data Scientists

| Need                    | GCP Service      | Why GCP Wins                         |
|-------------------------|------------------|--------------------------------------|
| Petabyte SQL analytics  | BigQuery         | Invented serverless analytics        |
| Container orchestration | GKE              | Google invented Kubernetes           |
| ML training at scale    | Vertex AI + TPUs | 3-5× better price/perf for TF/JAX    |
| Stream processing       | Dataflow (Beam)  | Most mature managed Beam runner      |
| Global SQL transactions | Cloud Spanner    | No true multi-region ACID equivalent |

# Transferability of Cloud Skills

- 80% of concepts transfer directly across providers
- Architecture patterns are identical: regions, compute, object storage, IAM, serverless
- Service names change; mental models do not
- Learn on GCP (most intuitive for data science)
- The skill is the architectural thinking, not the CLI syntax

### **KEY TAKEAWAY — Cloud Comparison**

Learn concepts on GCP (most intuitive for data science), but recognize that cloud knowledge is portable. 80% transfers directly to AWS and Azure. The skill is architectural thinking.

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# Lesson Summary

## Five Things to Remember

# The Netflix Answer

- How does Netflix serve 260 million subscribers? Everything we covered today:
- Regions & Zones — span multiple regions so Iowa failure doesn't affect Singapore
- Compute — recommendation algorithms on Compute Engine or Cloud Run
- Storage — compressed video in Cloud Storage buckets with CDN caching
- BigQuery — every play, pause, skip feeds into tables for recommendation queries
- IAM — hundreds of microservices with precisely scoped service accounts

# Five Things to Remember (1/2)

- ① GCP = Google's infrastructure, for rent — battle-tested at a scale that dwarfs any enterprise workload
- ② Choose Regions thoughtfully — latency, GDPR data-residency, and inter-region egress costs all matter
- ③ Pick the right compute tier — IaaS for full control, PaaS/CaaS for fast iteration, K8s for complex orchestration

# Five Things to Remember (2/2)

- ④ BigQuery is not a database — it's a serverless columnar query engine. Minimize columns scanned.
- ⑤ IAM is not optional — Least Privilege, Service Accounts, and audit logs are table stakes for any production data-science project
- Next Session (Day 13): AWS — S3 instead of GCS, EC2 instead of Compute Engine, Athena instead of BigQuery

## **KEY TAKEAWAY — Final Takeaway**

Cloud computing gives you access to the same infrastructure powering the world's largest tech companies. The skill is knowing which services to combine and how to manage them responsibly.

# Thank You

Day 12 Complete · Next: Day 13 — AWS

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